

Transcriptomics Training

Introduction to Illumina short-read
RNA sequencing, bioinformatics and data analysis

DTU Biosustain

Data Science Platform

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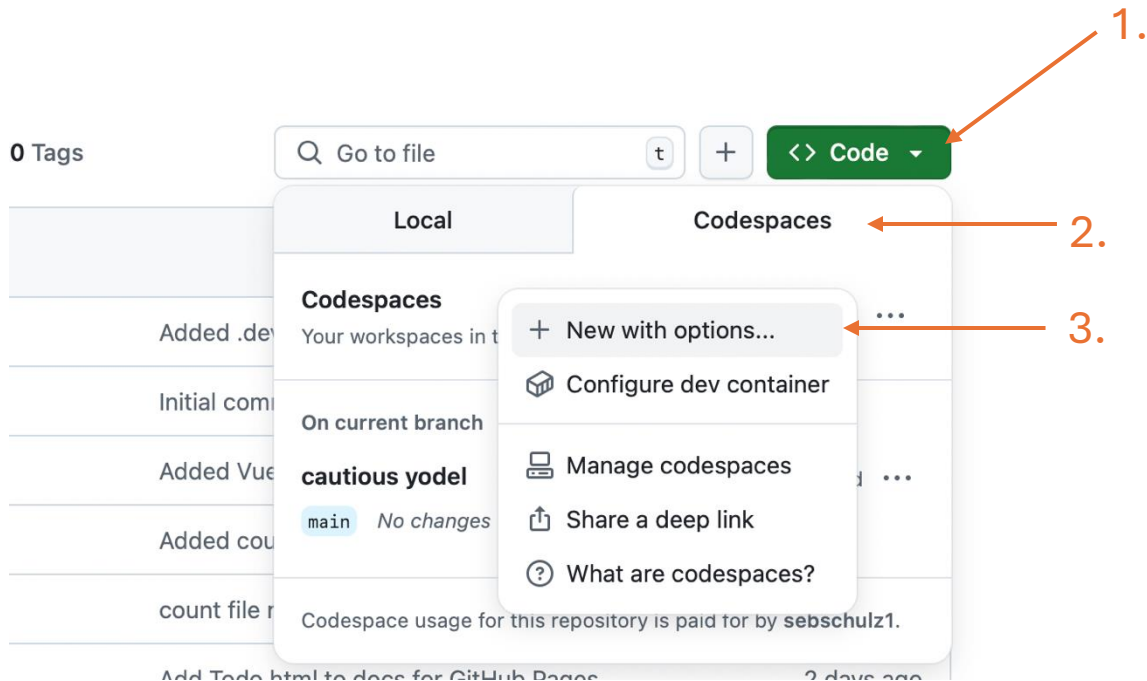
27 May 2025

Agenda – 8.30 – 16.30

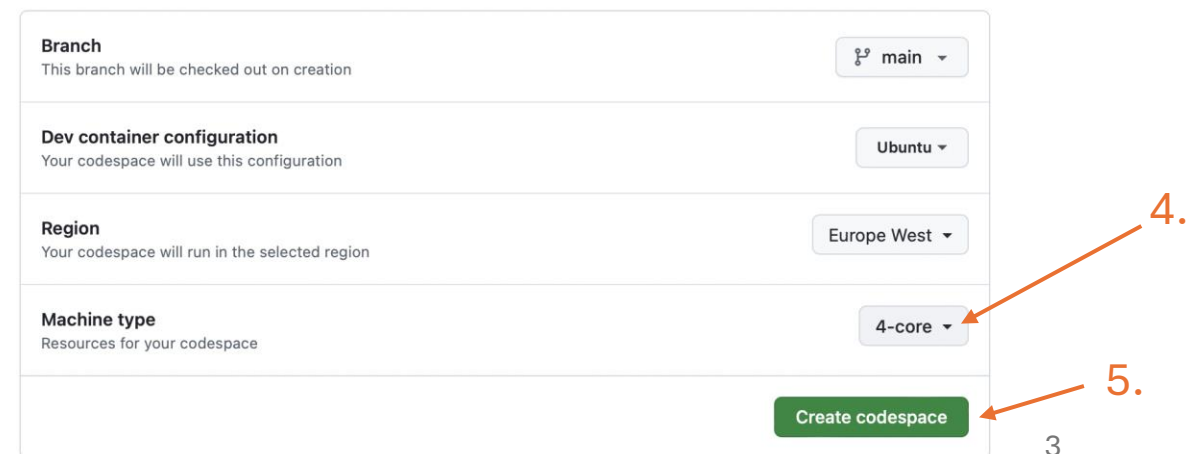
- **8.30:** Course introduction and coffee (Albert) – **30 min**
- **9.00: Session 1**
 - Theoretical background on RNAseq, incl. relevant molecular biology, and data processing (Sebastian) – **90 min** (25 slides)
- *10.30: Coffee break - 15 min*
- **10.45: Session 2**
 - Hands-on data processing using the nf-core/rnaseq pipeline (Sebastian) – **30 min** (*waiting time*)
 - Theoretical background on RNAseq data analysis (Sebastian) – **45 min** (14 slides)
- *12.00: Lunch – 30 min*
- **12.30: Session 3**
 - Results from data processing (Sebastian) – **30 min**
 - Hands-on RNAseq data analysis using R (Juliana) – **60 min**
- *14.00: Coffee break – 15 min*
- **14.15: Session 3 - continued**
 - Hands-on RNAseq data analysis using R (Juliana) – **60 min**
- **15.15: Session 4**
 - Results reporting using vuegen (Albert) – **45 min**
- *16.00-16.30: Wrap-up and final Q&A*

Create the codespace

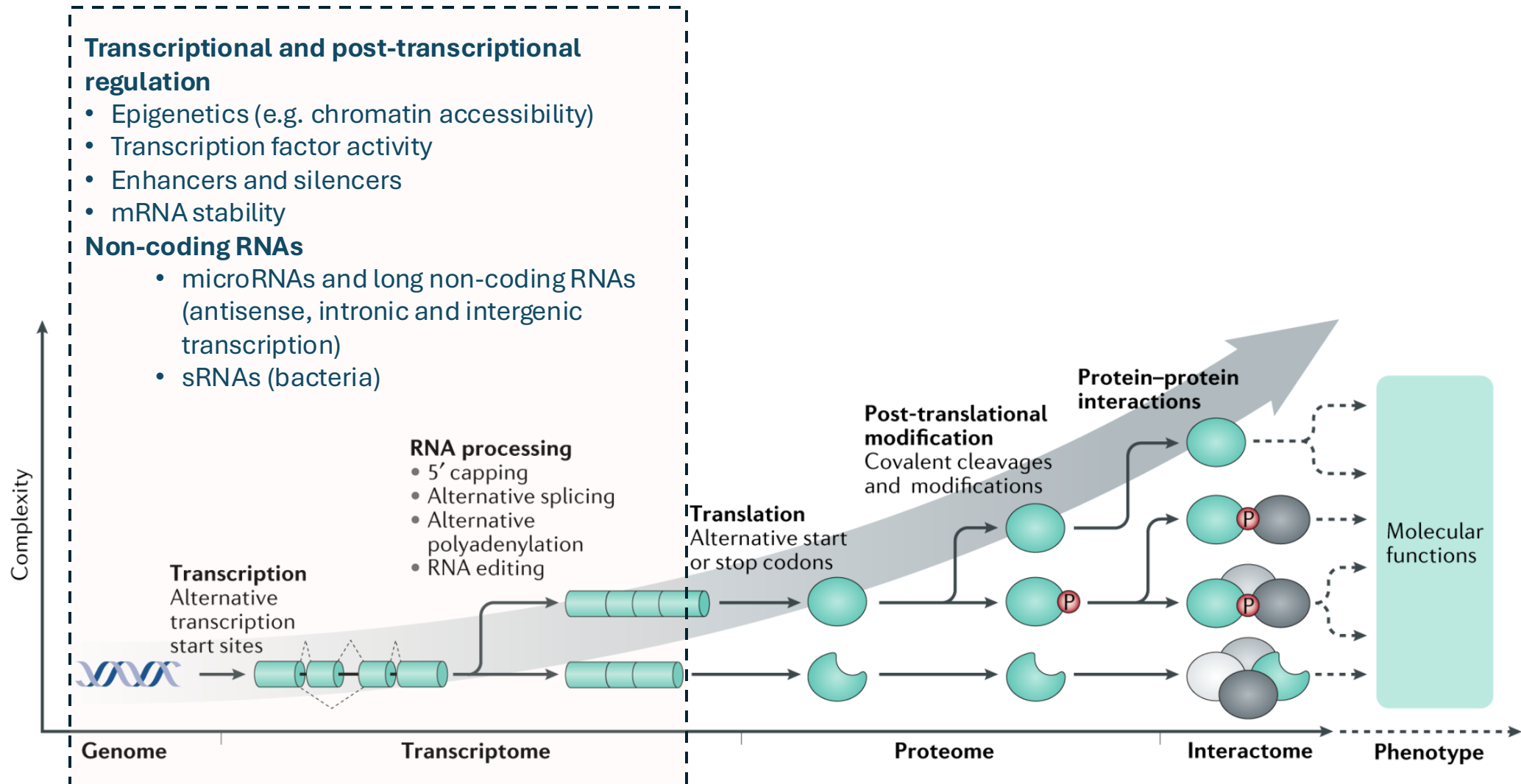
- Log into your github account
- Open the training repo: https://github.com/biosustain/dsp_transcriptomics_training
- Create codespace **with 4 cores** which we will use to run code



Select **4-cores** for machine type before creating the codespace



Studying the transcriptome



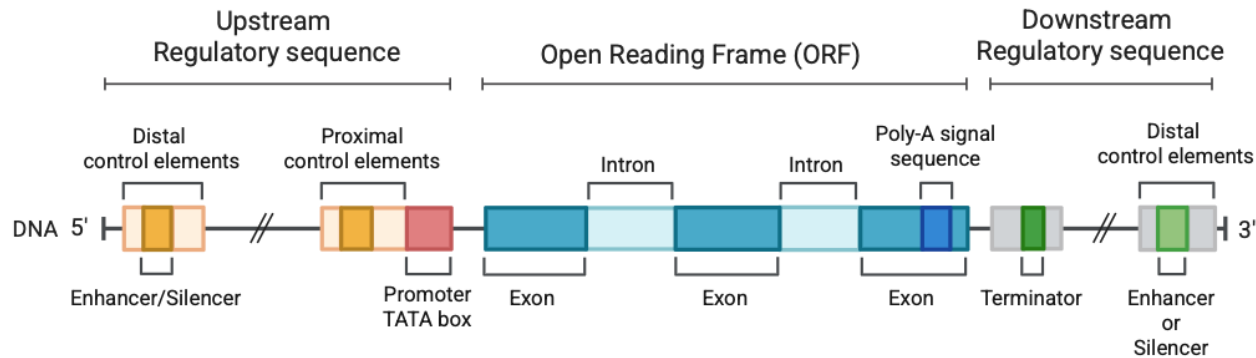
Eukaryotic example shown here

Modified from <https://doi.org/10.1038/s41580-020-0231-2>

Inspired by Marco Reverenna, Proteomics course 2025 ([Slides on github](#))

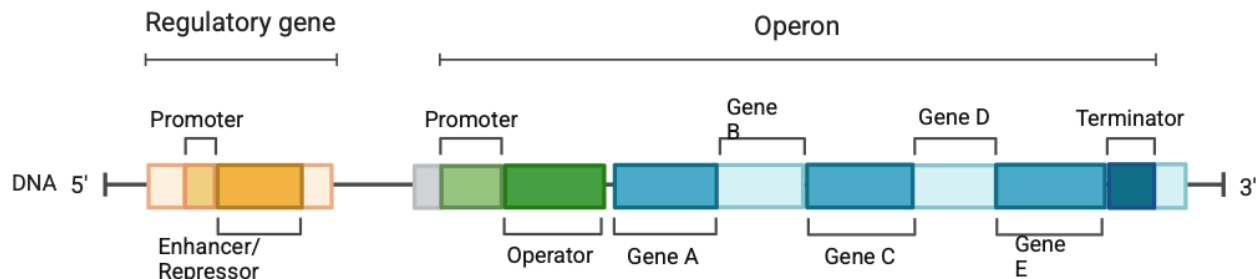
Eukaryotic and prokaryotic gene structure

Eukaryotic Gene Structure



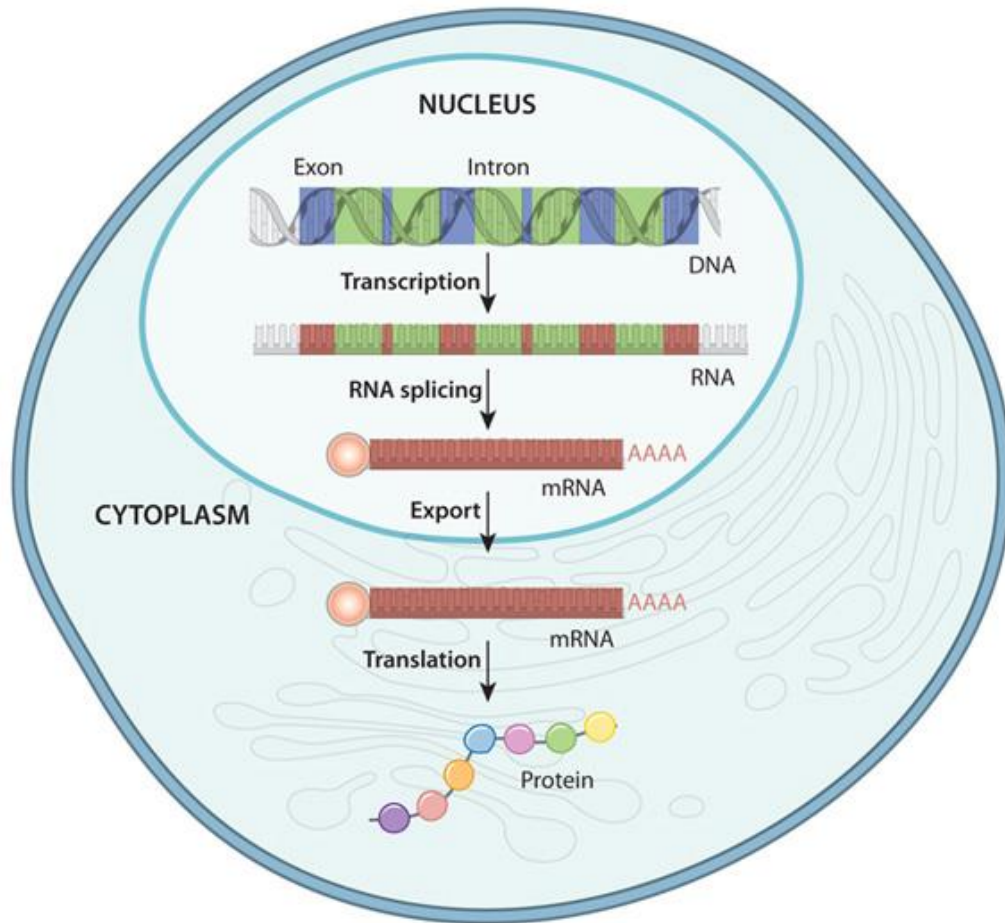
- Mostly **monocistronic**
- Exons and introns
- **Impact on data analysis:** align reads against exons only if the goal is to study gene expression of protein-coding genes (done in today's course)

Prokaryotic Gene Structure



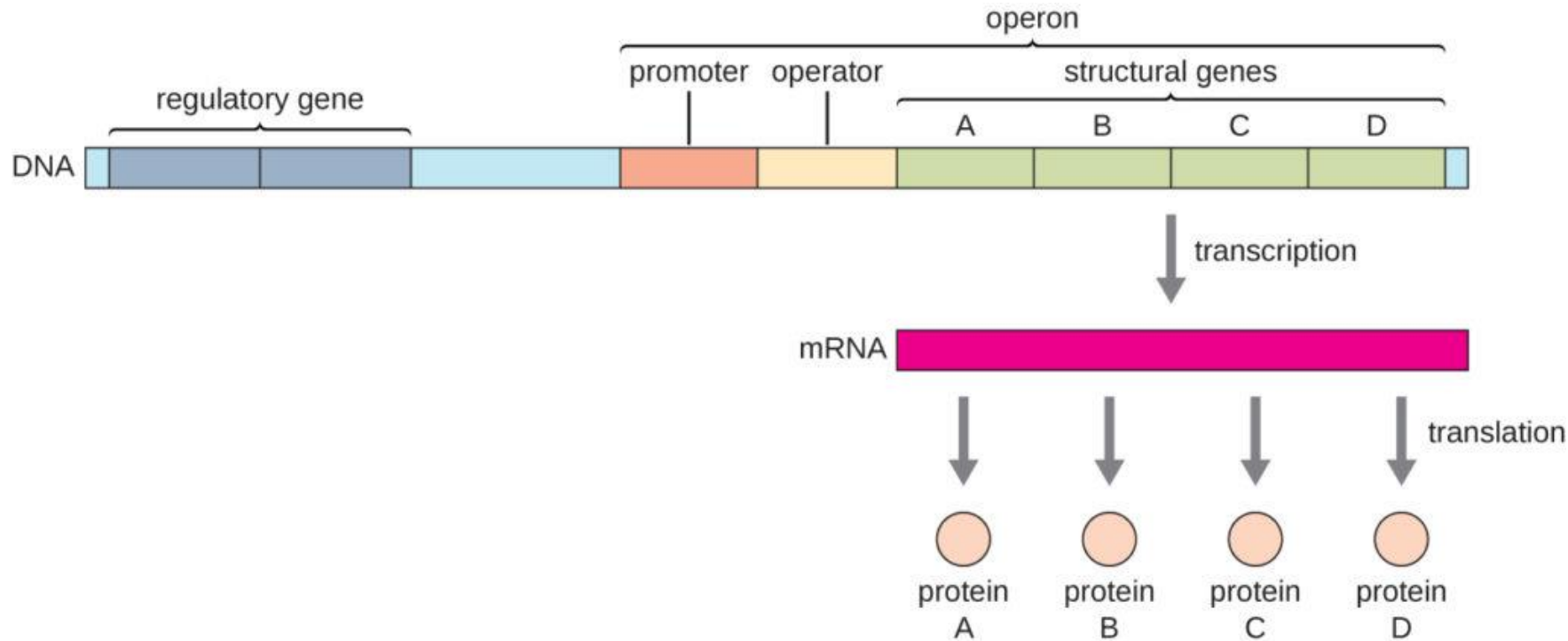
- Many **polycistronic**, i.e. many genes organised in an operon, but also **monocistronic**

Eukaryotic gene expression



- Usually monocistronic gene structure with exons and introns
- Eukaryotic mRNAs are poly-adenylated
- Pre-mRNA undergoes splicing to remove introns leaving exons that encode a protein (note there is also alternative splicing)

Prokaryotic gene expression



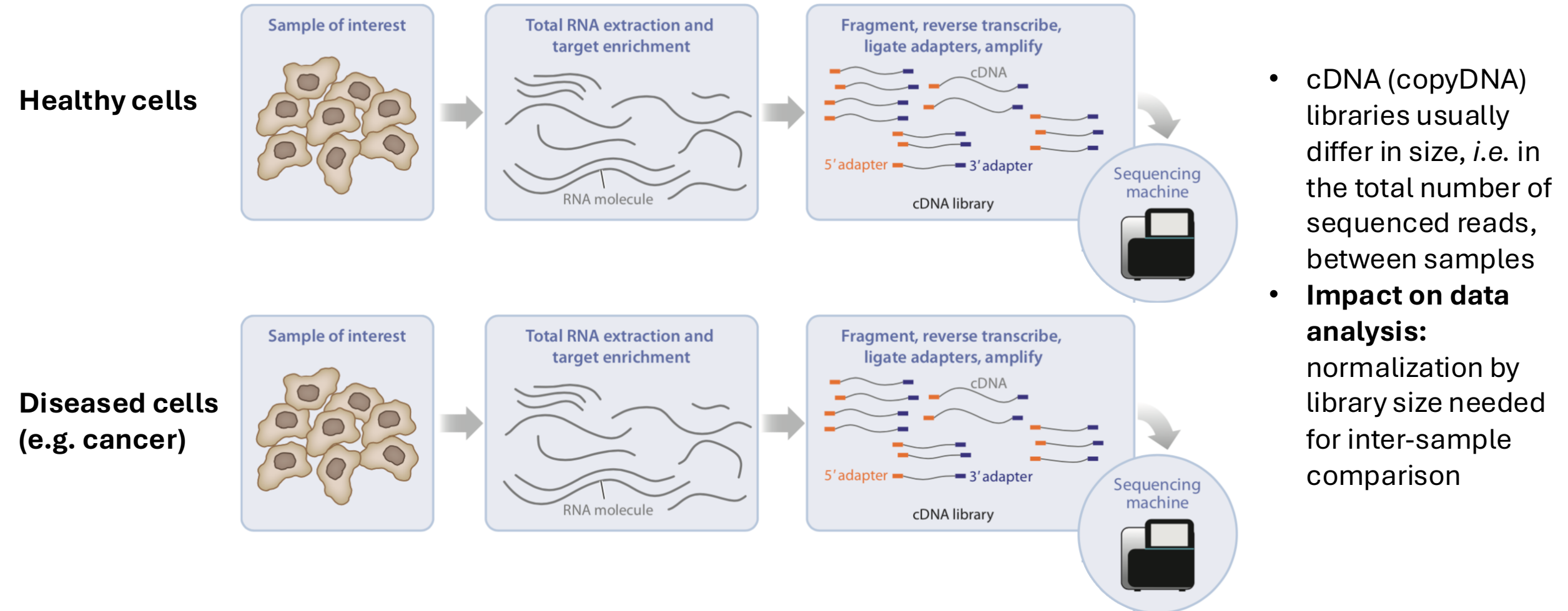
Polycistronic mRNA (encodes many ORFs encoding individual proteins)

Transcriptome analysis using RNA sequencing (RNAseq)

- **Short read sequencing (today's topic)**
 - **Illumina (50 – 300 base pairs)**
- Long-read sequencing
 - 1000 – 10000 base pairs
 - Oxford Nanopore
 - PacBio

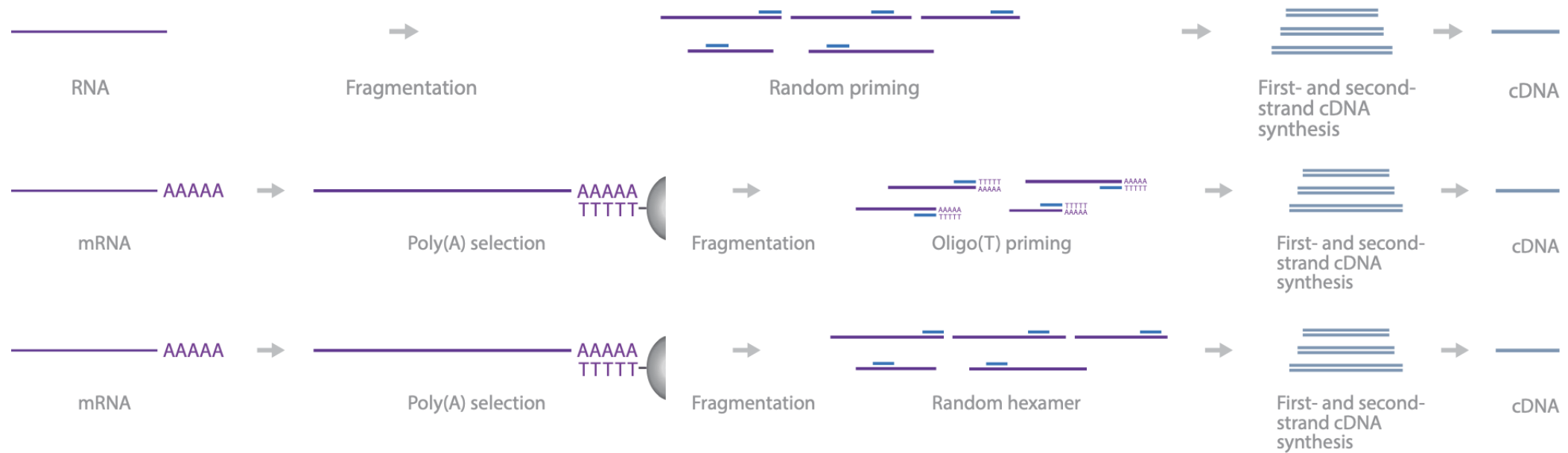
Illumina sequencing technology – RNAseq

- No direct RNA sequencing! Instead, RNA is enzymatically reverse transcribed to yield cDNA that is sequenced



Illumina sequencing technology – RNAseq

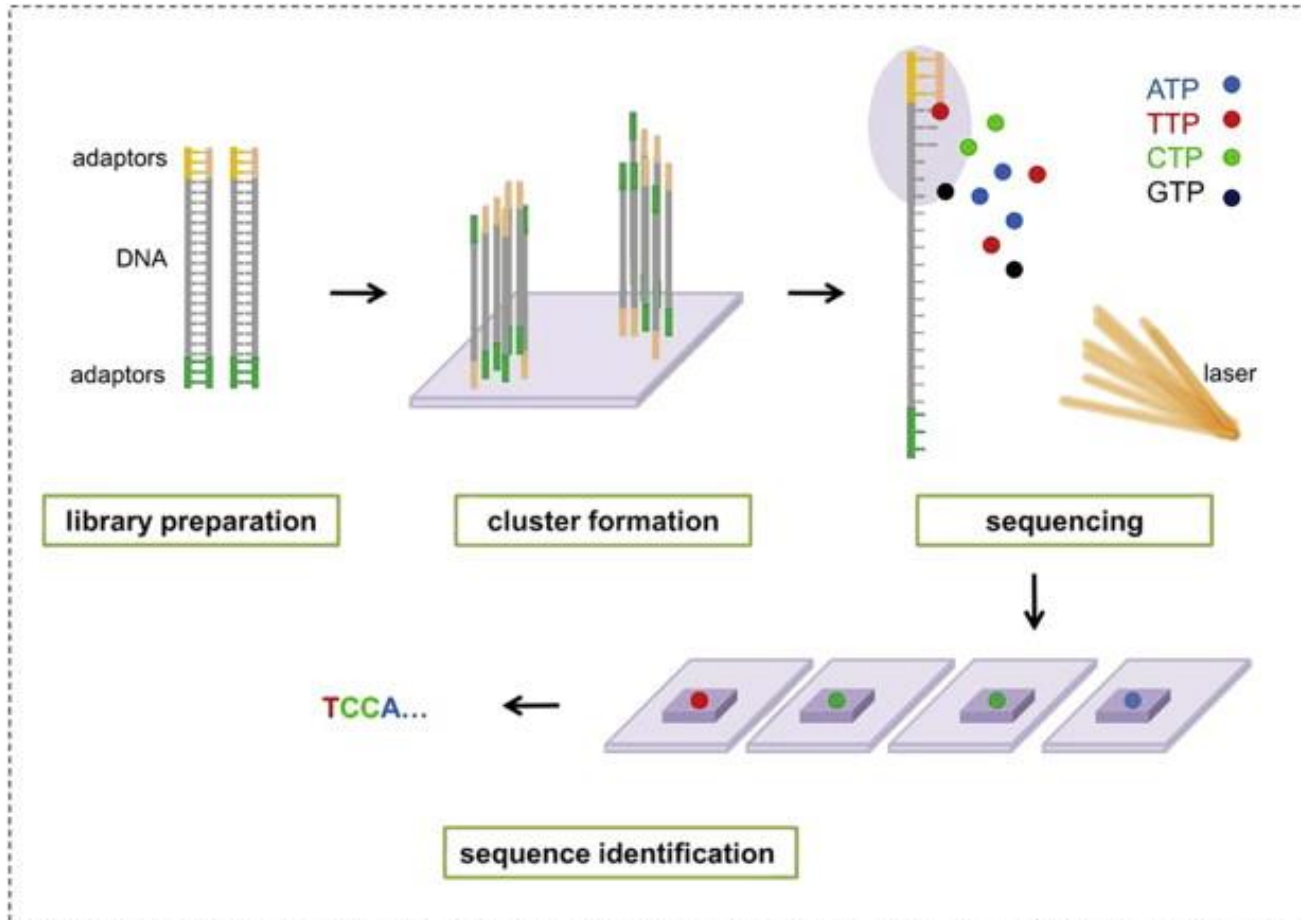
- Enrichment of poly-adenylated RNAs possible (target enrichment)
- Random or oligo(T) priming for reverse transcription



A schematic overview of RNA-Seq.

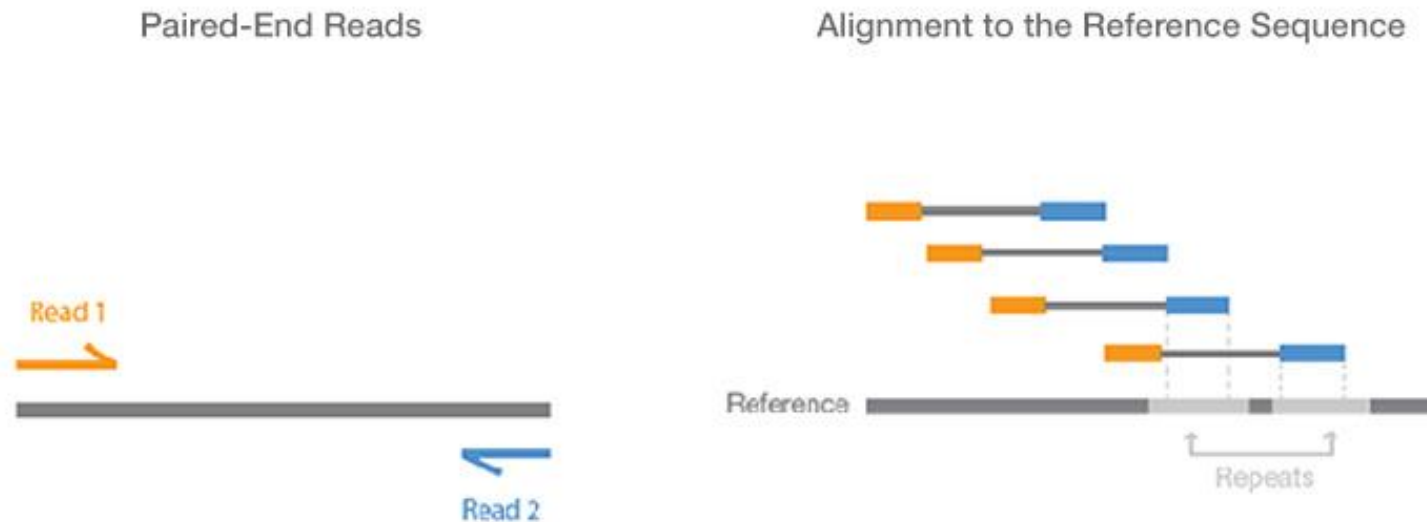
Illumina sequencing technology

- Massively parallel sequencing of DNA immobilized on a solid surface of a flow cell



- Adaptors are ligated to fragmented DNA
- DNA is attached on a surface and amplified forming clusters
- Sequencing-by-synthesis:
 - Incorporation of fluorophore-labeled dNTPs is measured during DNA synthesis
- Impact on data analysis:** adaptor sequences have to be removed prior to analysis (adaptor trimming)

Illumina sequencing technology – Paired-end sequencing

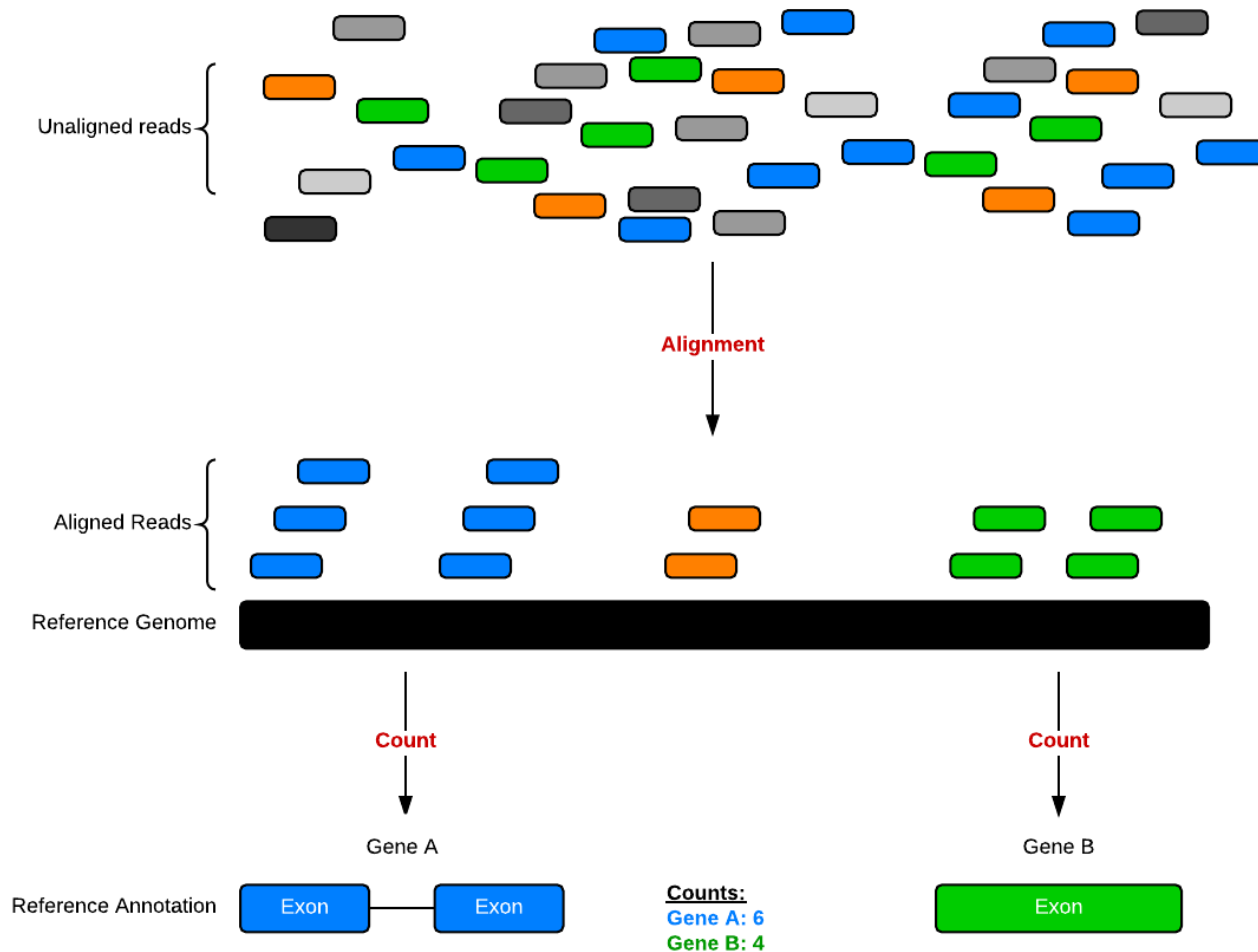


Paired-end sequencing enables both ends of the DNA fragment to be sequenced. Because the distance between each paired read is known, alignment algorithms can use this information to map the reads over repetitive regions more precisely. This results in much better alignment of the reads, especially across difficult-to-sequence, repetitive regions of the genome.

- Higher accuracy when DNA is sequenced from both ends
- Reads can be aligned with higher precision, especially in complex genomic regions (repetitive sequences)
- Detection of genomic rearrangements
- Detection of gene fusions and novel transcripts

[Source](#)

Major steps and goals in RNAseq



Major steps in RNAseq analysis

- Quality control (QC)
- Alignment: map reads to a reference sequence (e.g. genome)
- Read quantification: count how many reads are aligned to a specific genomic region (feature)

Focus in this course is quantification

- One major goal of RNAseq is differential expression (DE) analysis comparing read counts of a specific genomic feature between biological experiments

Healthy cells

Gene

—
—

Cancer cells

Gene

—
—
—
—
—

Further goals (not covered today)

- New transcript detection
- New isoform detection (alternative splicing)

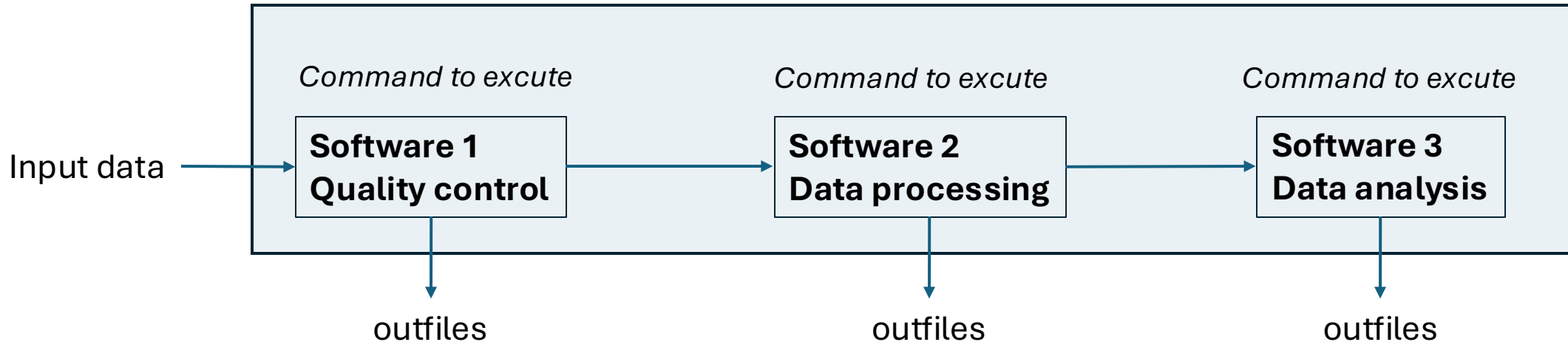
Bioinformatics for Illumina short read data

- **1. Data processing (nf-core/rnaseq pipeline) – selected softwares**
 - Quality control (QC), MultiQC
 - Read alignment
 - Read quantification
- **2. Post-processing data analysis**
 - Normalization methods
 - Differential gene expression (DGE *or* DE) analysis
 - Statistics
 - DESeq2 (R language)
 - Functional analysis for biological interpretation
 - Gene-set enrichment analysis
- **3. Results reporting using vuegen**

1. Data processing – pipelining with nextflow

- RNAseq data analysis requires the consecutive execution of individual softwares
- Executing softwares one by one using the command line is possible but requires many manual steps
- Reproducible data processing and analysis can be achieved by chaining individual software versions using a dedicated pipelining tool, e.g. nextflow
- Execution of pipeline is done using a single command
- Nextflow: Portable (across different platforms), scalable (parallel data processing), reproducible (standardized workflow)

Pipeline: Single command to run the pipeline



1. Data processing – nf-core



The nf-core framework for community-curated bioinformatics pipelines

[Philip A. Ewels](#), [Alexander Peltzer](#), [Sven Fillinger](#), [Harshil Patel](#), [Johannes Alneberg](#), [Andreas Wilm](#),
[Maxime Ulysse Garcia](#), [Paolo Di Tommaso](#) & [Sven Nahnsen](#) 

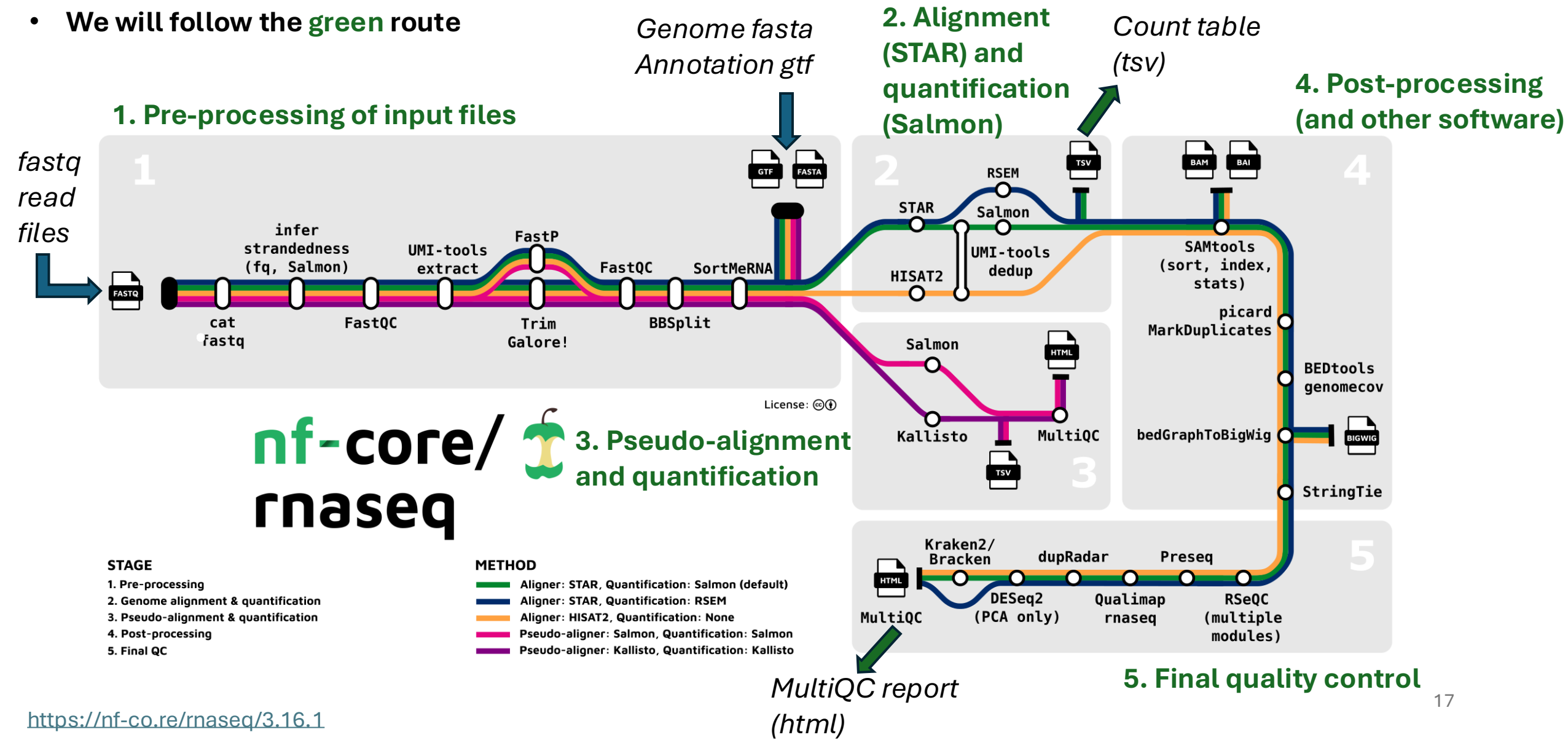
[Nature Biotechnology](#) **38**, 276–278 (2020) | [Cite this article](#)

34k Accesses | **166** Altmetric | [Metrics](#)

- **Collection of nextflow pipelines for bioinformatics**
 - 130 pipelines available (80 released, 38 under development, 12 archived)
 - Standardized and parallel data processing (e.g. in the cloud)
 - Pipelines undergo rigorous testing before release
 - Continuously tested and maintained pipelines
 - Direction of development is heavily discussed with a community of developers
 - Active community of developers and users with fast support and troubleshooting
 - Regular world-wide hackathons

1. Data processing – nf-core/rnaseq pipeline

- We will follow the **green** route



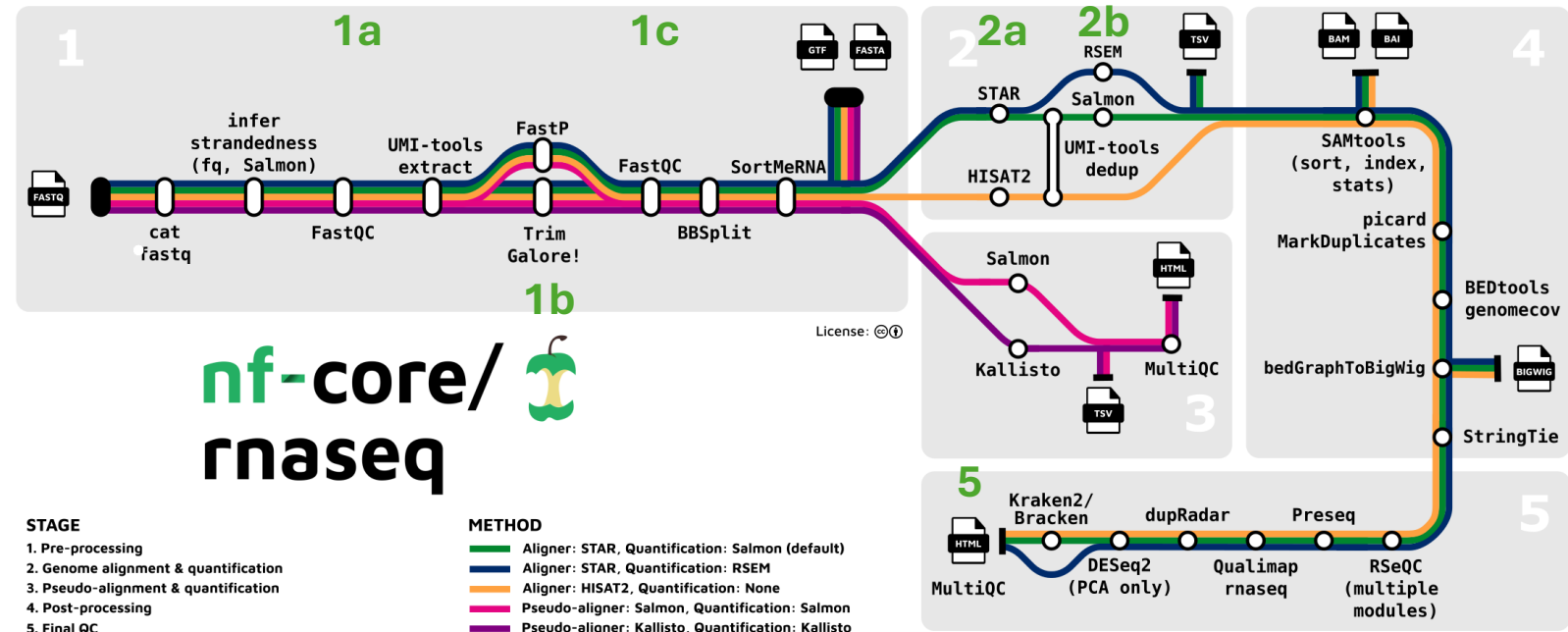
1. Data processing – nf-core/rnaseq pipeline

Overview of selected individual steps

- 1a) **Quality control (QC)** with **FastQC** of sequencing reads
- 1b) **Adaptor trimming** with **Trim Galore** to remove adaptor sequences that were ligated to DNA prior to sequencing
- 1c) **FastQC** again
- 2a) Read alignment with **STAR** to map of reads against a reference sequence (e.g. genome sequence)
- 2b) Read quantification with **Salmon** to quantify how many reads were mapped to a specific genomic feature (e.g. gene)
- ...
- 5) **MultiQC** report consolidates all quality control results obtained from several softwares

Input files

- Sequencing files (.fastq)
- Genome sequence (.fasta)
- Genome annotation (.gtf)
- (Optional additional files)



1. Data processing input– sequencing files (.fastq)

- Fastq files contain information about the sequenced reads (one example entry shown below)
- One entry has 4 lines

```
@ML-P2-14:9:000H003HG:1:11102:17290:1073 1:N:0:TCCTGAGC+GCGATCTA  
TTTGGTAACAGCATGAATTATTCTAGCCACTAAACTCTATGAACATCTTGTGAAGGTTTCAGATAGAGCCTGAAGTACACAGAGAACAATTCTTAAAAA  
+  
AAAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE<AEEEEEEE
```

Line 1: A sequence identifier with information about the sequencing run and the cluster.

Line 2: DNA sequence (the base calls; A, C, T, G and N).

Line 3: A separator, a plus (+) sign.

Line 4: Base call quality scores (Phred +33 encoded, using ASCII characters to represent the numerical quality scores).

1. Data processing input – genome sequence file (.fasta, .fna)

- DNA sequence for each chromosome as fasta entry (starting with “>”)
- Each fasta entry has a header and the DNA sequence of the respective chromosome
- Example for mouse genome fasta file entries shown below (sequences are putative)

```
>1 dna_rm:chromosome chromosome:GRCm39:1:1:195154279:1 REF  
ATCGTA.....
```

```
>2 dna_rm:chromosome chromosome:GRCm39:2:1:181755017:1 REF  
TTTGCA.....
```

```
>3 dna_rm:chromosome chromosome:GRCm39:3:1:159745316:1 REF  
TGGCA....
```

1. Data processing input – annotation file (.gtf)

- General Transfer Format (GTF) with nine columns
- Contains information about annotated sequence features (e.g. gene, exon, etc)
- Snapshot of mouse gtf file shown below

```
4   havana   CDS   98592232   98592338   .   +   0   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; transcript_vers
4   havana   exon   98598021   98598096   .   +   .   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; transcript_
4   havana   CDS   98598021   98598096   .   +   1   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; transcript_vers
4   havana   exon   98606146   98607840   .   +   .   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; transcript_
4   havana   CDS   98606146   98606166   .   +   0   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; transcript_vers
4   havana   stop_codon 98606167   98606169   .   +   0   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; transcr
4   havana   five_prime_utr 98284028   98284161   .   +   .   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; tra
4   havana   five_prime_utr 98293616   98293649   .   +   .   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; tra
4   havana   three_prime_utr 98606170   98607840   .   +   .   gene_id "ENSMUSG00000061859"; gene_version "18"; transcript_id "ENSMUST00000238306"; tra
```

1	2	3	4	5	6	7	8	9
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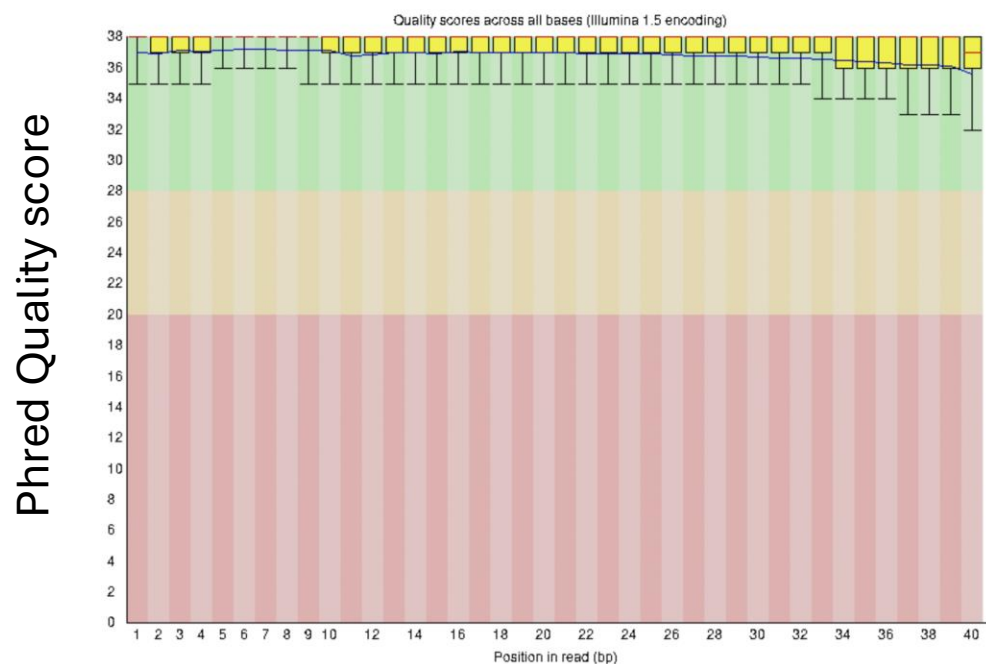
1. **seqname:** name of the chromosome or scaffold
2. **source:** name of the program that generated this feature, or the data source (database or project name)
3. **feature:** feature type name, e.g. gene, exon
4. **start:** start position* of the feature, with sequence numbering starting at 1.
5. **end:** end position* of the feature, with sequence numbering starting at 1.
6. **score:** a floating point value.
7. **strand:** defined as + (forward) or - (reverse).
8. **frame:** one of '0', '1' or '2'. '0' indicates that the first base of the feature is the first base of a codon, '1' that the second base is the first base of a codon, and so on..
9. **attribute:** a semicolon-separated list of tag-value pairs, providing additional information about each feature.

1. Data processing – Quality control with FastQC

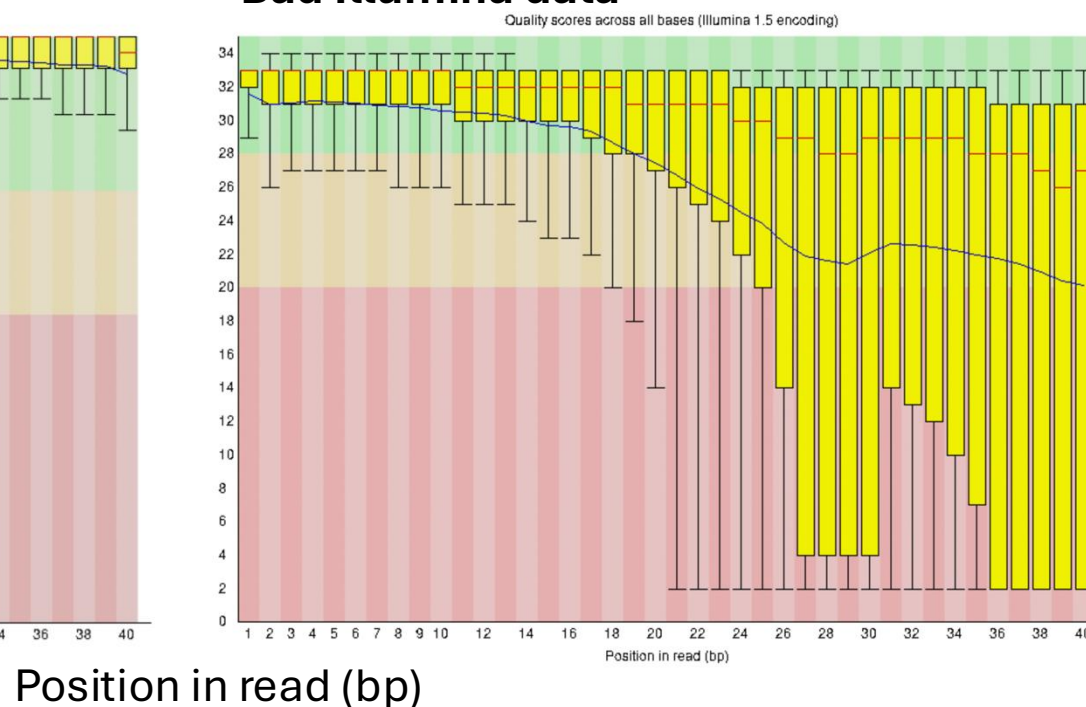
- FastQC assesses the quality per base pair position of sequencing reads and assigns a Phred quality score
- Assesses also read length distribution as well as per sequence GC, N and adaptor content
- Conducted twice: before and after adapter trimming

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%

Good Illumina data



Bad Illumina data

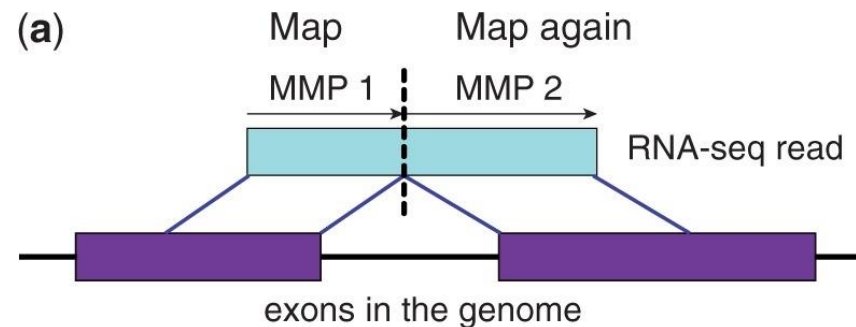


1. Data processing - Read alignment with STAR

- Spliced Transcripts Alignment to a Reference (STAR)
- Very fast aligner
- Alignment against genome (or transcriptome)
- *De novo* detection of splice junctions (no prior annotation of splicing event required)

Reference genome

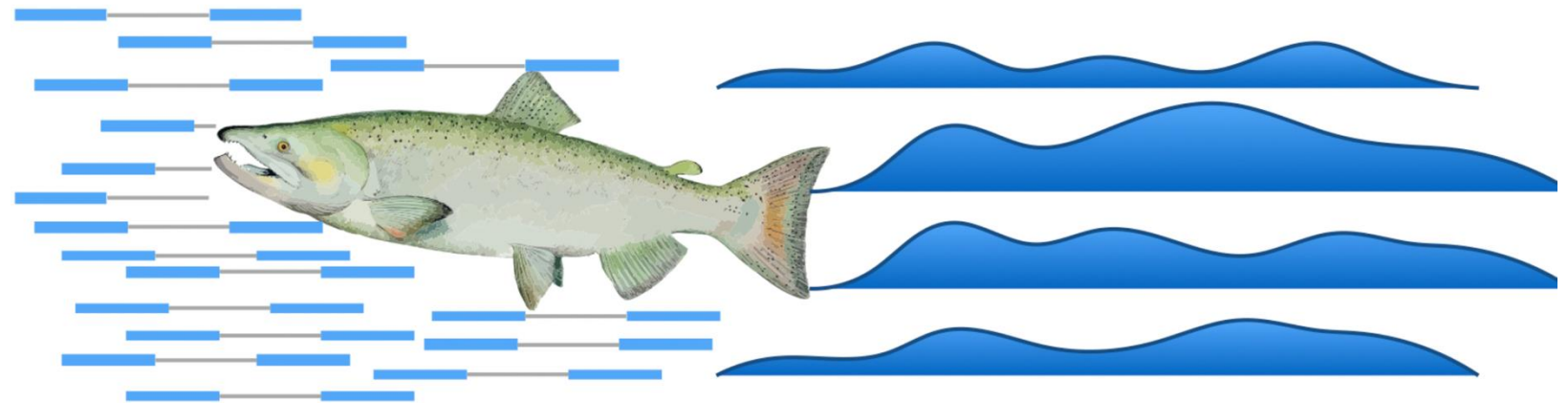
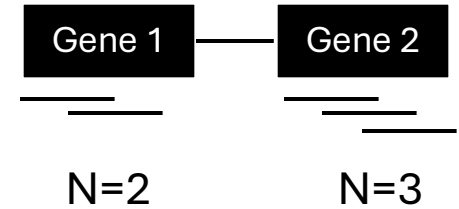
Detection of splice junctions with STAR



MMP: Maximal Mappable Prefix

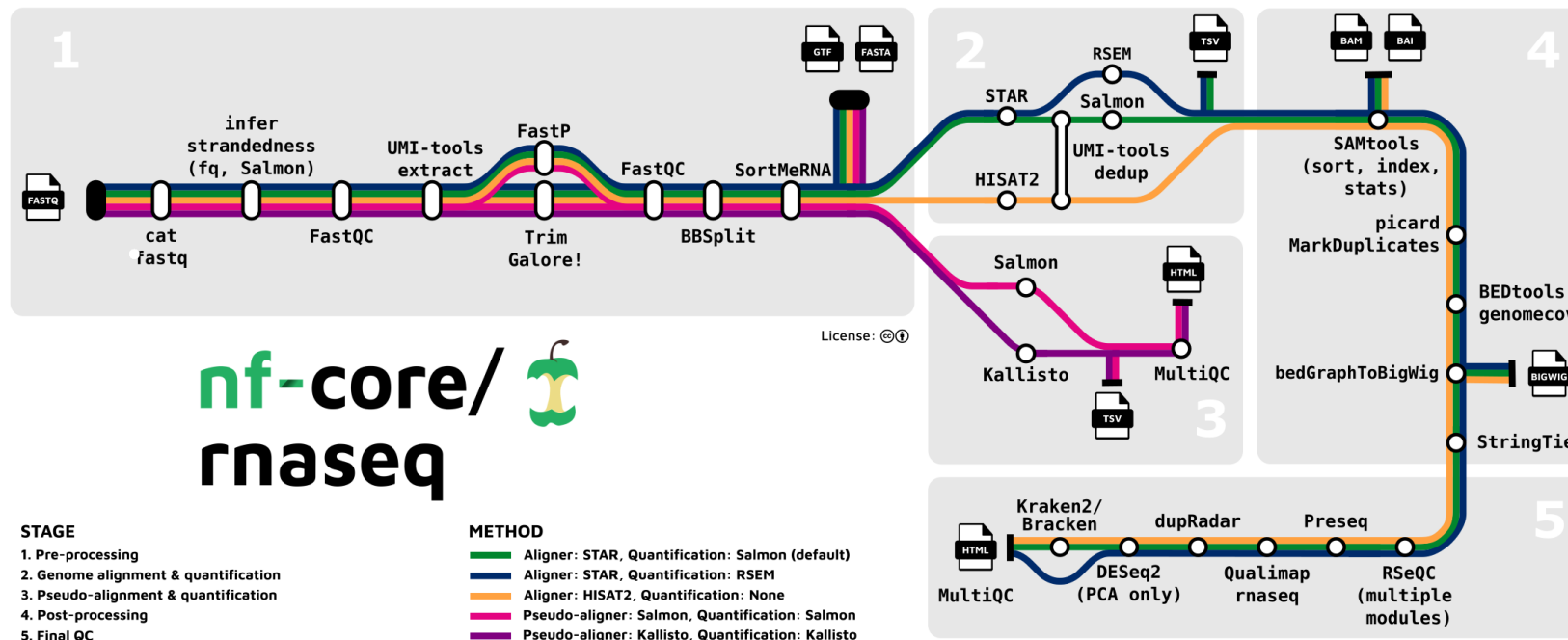
1. Data processing - Read quantification with Salmon

- More complex quantification procedure compared to other read counters/quantifiers
- Handles counting of multi-mapping reads
- More accurate quantification of reads achieved by considering sample-specific parameters and biases of RNAseq data:
 - positional biases in coverage
 - sequence-specific biases at the 5' and 3' ends of sequenced fragments
 - fragment-level GC bias
 - strand-specific protocols
 - fragment length distribution



Hands-on data processing with nf-core/rnaseq

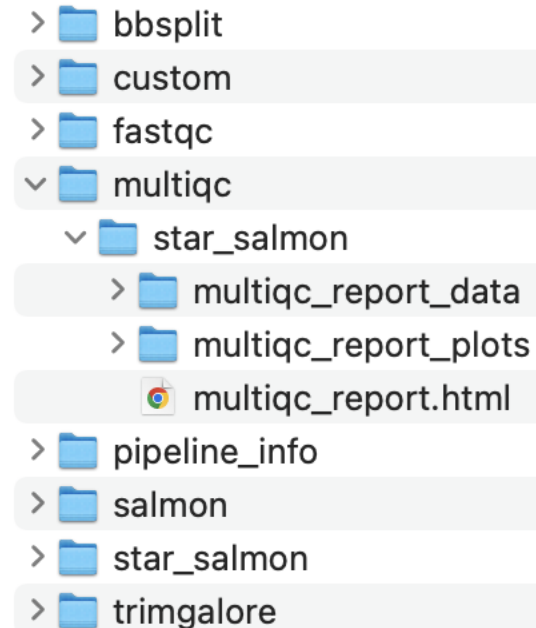
- Example nextflow command (not executable):
 - https://github.com/biosustain/dsp_transcriptomics_training/blob/main/nf_core_launch/nf_command_example.sh
- Due to long processing time, we will run a test data set that comes with the pipeline itself
 - Instructions :
https://github.com/biosustain/dsp_transcriptomics_training/blob/main/course_contents/launching_rnaseq.md



Lunch break

1. Data processing – output files

- nf-core pipeline outputs are folders from the different softwares executed
- **Multi-QC report (html) examples:**
 - General stats
 - STAR alignment stats
 - FastQC(raw)
 - Sequence Quality Histograms
 - Per Sequence GC Content
 - Per Base N Content
 - STAR
 - SALMON
 - DESeq2 PCA
 - Software versions
 - nf-core/rnaseq Workflow Summary
- **Star_salmon folder:**
 - Count tables: read counts per sequence feature



```
> bbsplit
> custom
> fastqc
v multiqc
  v star_salmon
    > multiqc_report_data
    > multiqc_report_plots
    multiqc_report.html
> pipeline_info
> salmon
> star_salmon
> trimgalore
```

2. Post-processing data analysis

- Normalization methods
- Differential gene expression (DGE *or* DE) analysis
 - Statistics
 - DESeq2 (R language)
- Functional analysis for biological interpretation
 - Gene-set enrichment analysis

Normalization methods

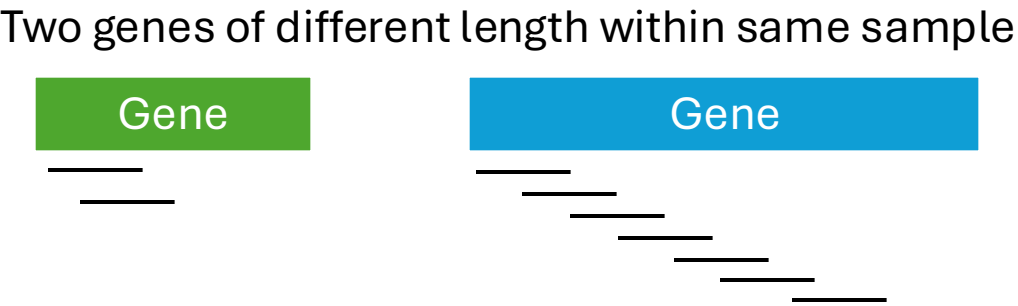
Read counts per gene depend on (among others):

- 1. Library size (count of total reads)
- 2. Gene length
- 3. RNA composition

1. Library size



2. Gene length

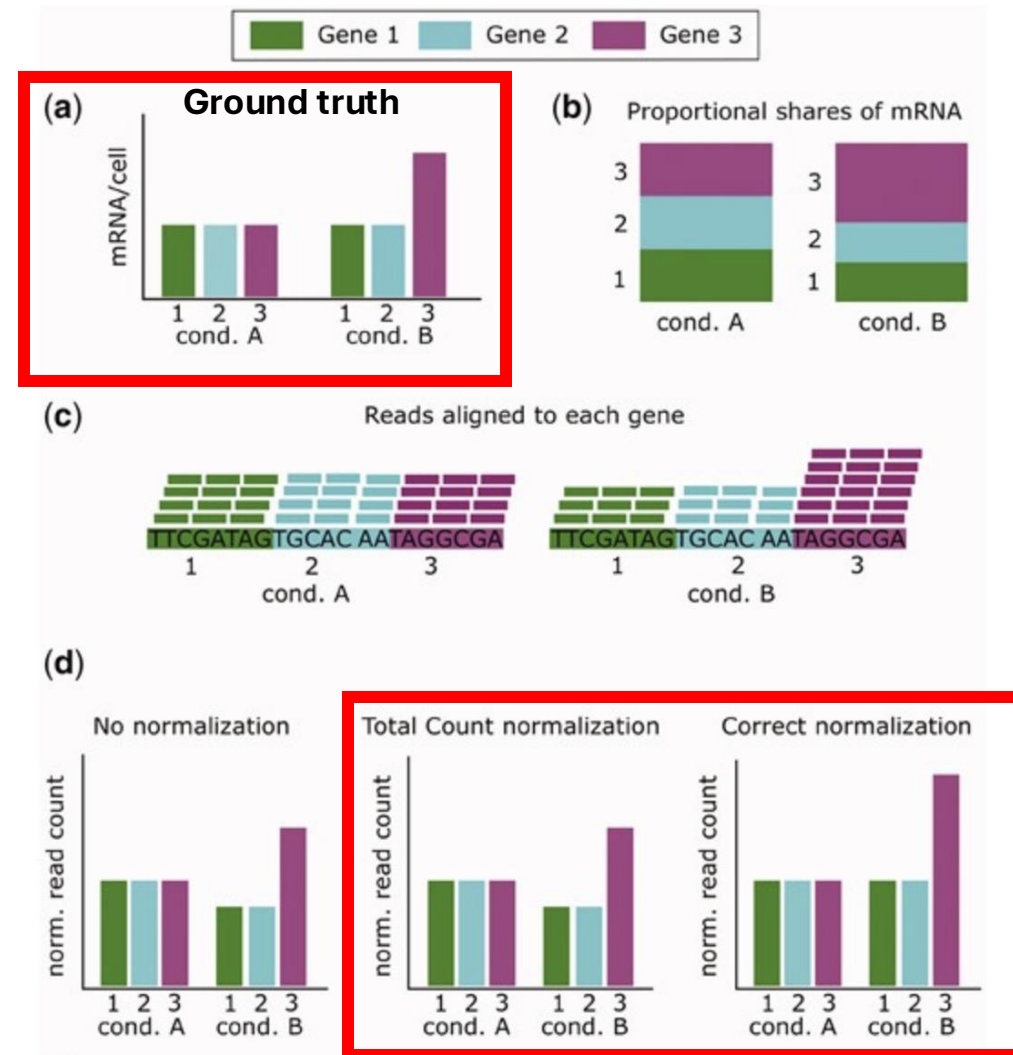
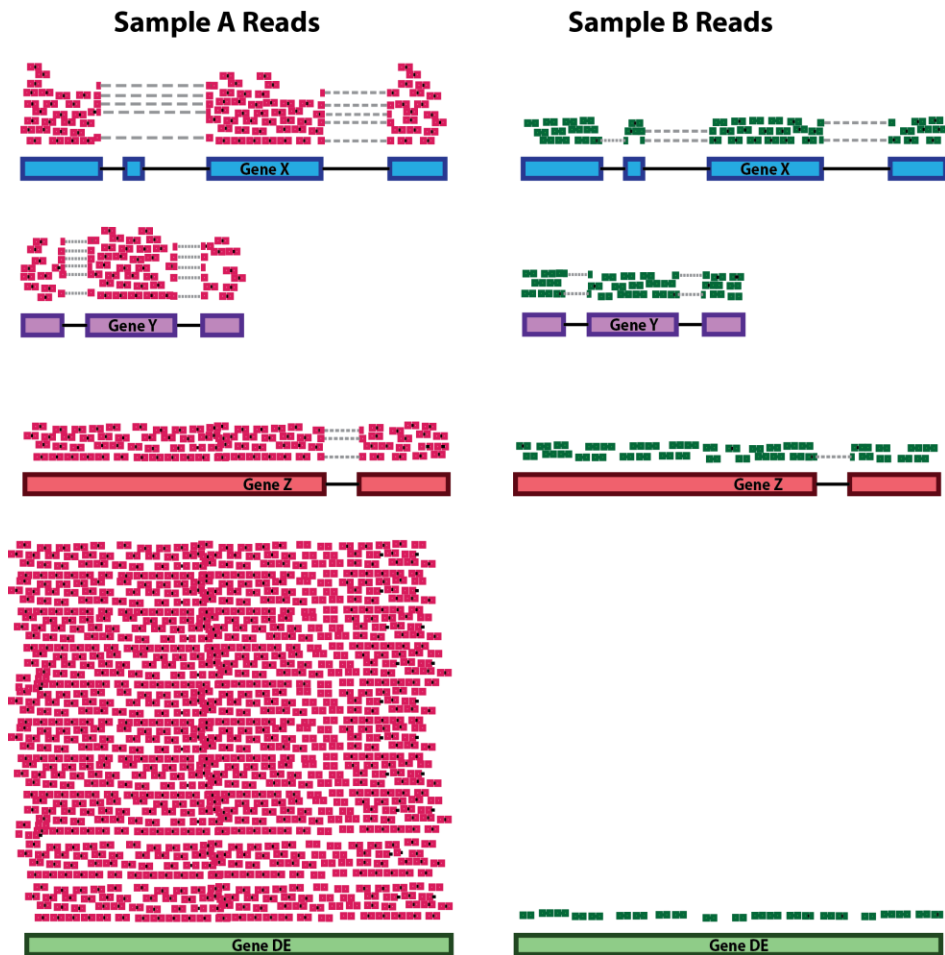


Read count	2	4	2	8
Count of total reads	n	2n	n	n
Gene length	L	L	L	4L
Normalized count	2	2	2	2
Relevant for	Inter-sample comparison		Intra-sample comparison	

Normalization methods

Read counts per gene depend on (among others):

3. RNA composition



- Many reads for a single highly expressed gene can make genes expressed at lower levels appear to be even lower
- This doesn't reflect the ground truth (see figure a)

Normalization methods – selection

Method	Description	Library size normalized	Gene length normalized	RNA composition normalized	Useful for comparisons	Comment	Refs
CPM	Counts per million mapped reads	yes	no	no	Inter-sample of same sample group		
RPKM/FPKM	Reads/fragments per kilobase per million mapped reads	yes	yes	no	Intra-sample	Not so much in use anymore	
TPM	Transcripts per million	yes	yes	no	Intra-sample; inter-sample of same sample group	Sum of gene TPM values is the same across samples	Link
TMM	Trimmed mean of M values	yes	no	yes	Inter-sample and quantitative DE analysis	Used in edgeR package	Link
DEseqCounts	Median of the ratios	yes	no	yes	Inter-sample and quantitative DE analysis	Used in DESeq2 package	Link

- Choice of a normalization method can depend on several factors, e.g. the downstream analysis
- DESeq2 is one of the standards for quantitative estimation of differential gene expression

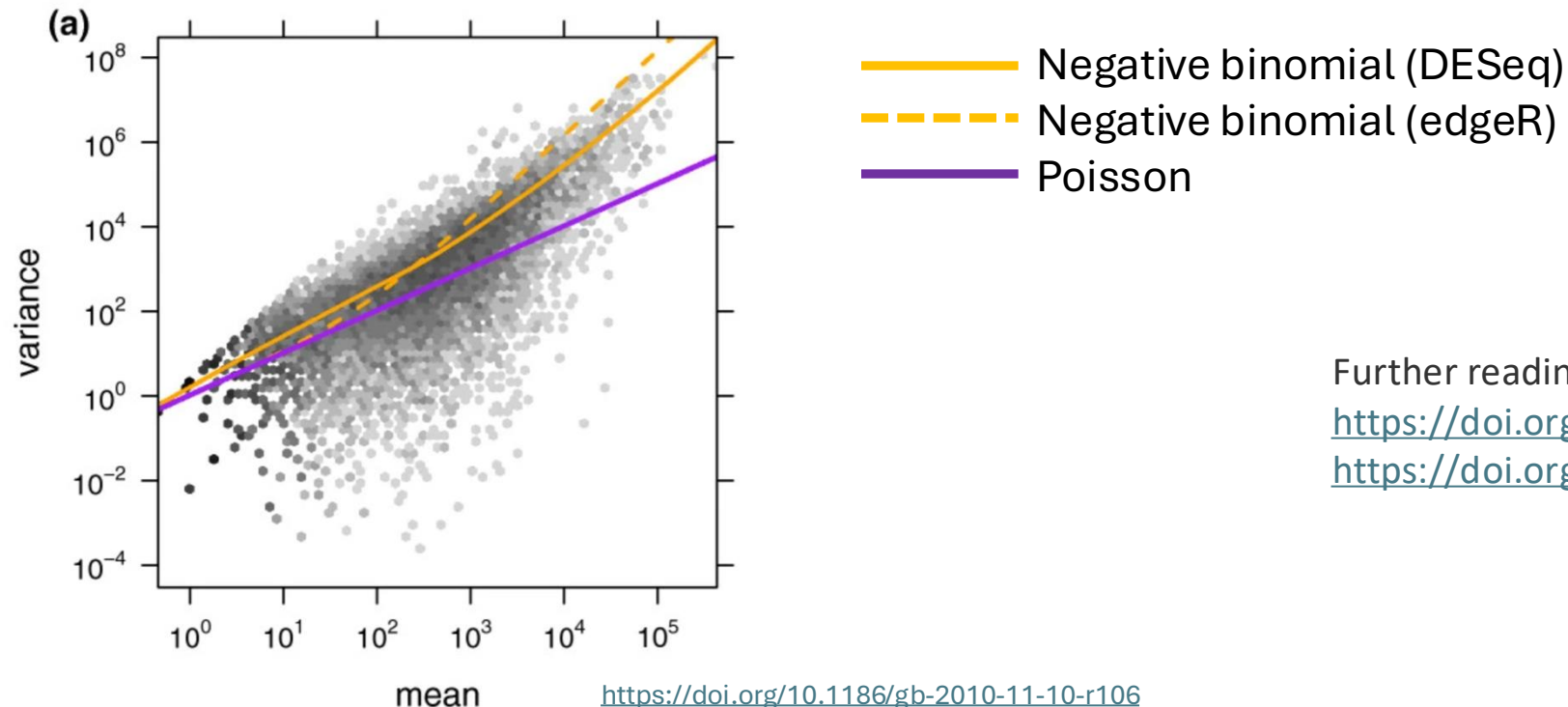
[Source link](#)

Differential gene expression analysis with DESeq2

- Main goal of differential gene expression analysis is to determine genes that are significantly different expressed between experimental conditions
 - Effect size: calculation of **fold-change** or **log2-fold-change (LFC)**
 - Statistical inference: calculation of **p-values adjusted for multiple testing**
- DESeq2 and its predecessor DESeq are R libraries specifically designed for the statistical analysis of bulk RNAseq count data
 - **Important:** never use normalized reads as input to DESeq2!!!
 - DESeq2 accepts only raw read counts and uses the on the previous slide described internal count normalization!!!
- Statistical model of DESeq2
 - RNAseq data distribution
 - Normalization (DESeqCounts)
- Hands-on

Differential gene expression analysis – data distribution

- RNAseq read counts have to be modelled by a discrete probability distribution
- Initially, Poisson distribution (mean=variance) was used, but this did not account for the observed overdispersion between biological replicates (counts from biological replicates have higher variance than the mean)
- RNAseq data can be better modelled using the **negative binomial distribution (NB)**; R packages DESeq2 and edgeR use NB distribution models
- Information of data distribution is important for statistical inference from a limited number of experimental replicates



Further reading: DESeq and DESeq2 papers
<https://doi.org/10.1186/gb-2010-11-10-r106>
<https://doi.org/10.1186/s13059-014-0550-8>

Differential gene expression analysis – details of the DESeq2 model

- Assumption: most genes are not differentially expressed
- Size factors s_j are incorporated into the negative binomial model to allow normalization by library size and RNA composition
- Log2-fold changes (LFC), e.g. change of genes between untreated and treated condition, are obtained from a generalized linear model

Count matrix K with $K_{i,j}$ sequencing reads

gene	sample_1	sample_2	sample...	sample_j
gene_A	count	count	count	count
gene_B	count	count	count	count
gene_C	count	count	count	count
gene_...	count	count	count	count
gene_i	count	count	count	count

Counts K_{ij} for gene i , sample j are modelled using a negative binomial distribution (NB)

$$K_{ij} \sim \text{NB}(\mu_{ij}, \sigma_{ij}^2)$$

Mean scaled by **size factor** (normalization)

$$\mu_{ij} = s_j q_{ij}$$

LFC calculation using generalized linear model

$$\log_2(q_{ij}) = x_j \cdot \beta_i$$

μ_{ij} : mean

σ_{ij}^2 : variance

s_j : sample-specific size factor

q_{ij} : quantity proportional to the concentration of cDNA fragments from the gene in the sample

β_i : log2 fold change for gene i for each column of the model matrix X (e.g. treated or untreated sample)

DESeq2 normalization – DESeqCounts (size factors)

1. Create a pseudo-reference sample (row-wise geometric mean)

gene	sampleA	sampleB	pseudo-reference sample
EF2A	1489	906	$\sqrt{1489 * 906} = \mathbf{1161.5}$
ABCD1	22	13	$\sqrt{22 * 13} = \mathbf{17.7}$

2. Calculate ratio of each sample to the reference

gene	sampleA	sampleB	pseudo-reference sample	ratio of sampleA/ref	ratio of sampleB/ref
EF2A	1489	906	1161.5	$1489/1161.5 = \mathbf{1.28}$	$906/1161.5 = \mathbf{0.78}$
ABCD1	22	13	16.9	$22/16.9 = \mathbf{1.30}$	$13/16.9 = \mathbf{0.77}$

3. Calculate the normalization factor for each sample (size factor)

```
normalization_factor_sampleA <- median(c(1.28, 1.30))
```

```
normalization_factor_sampleB <- median(c(0.78, 0.77))
```

4. Calculate the normalized count values using the size factor

sampleA **median ratio** ~ 1.3
sampleB **median ratio** ~ 0.77

Raw Counts

gene	sampleA	sampleB
EF2A	1489	906
ABCD1	22	13

Normalized Counts

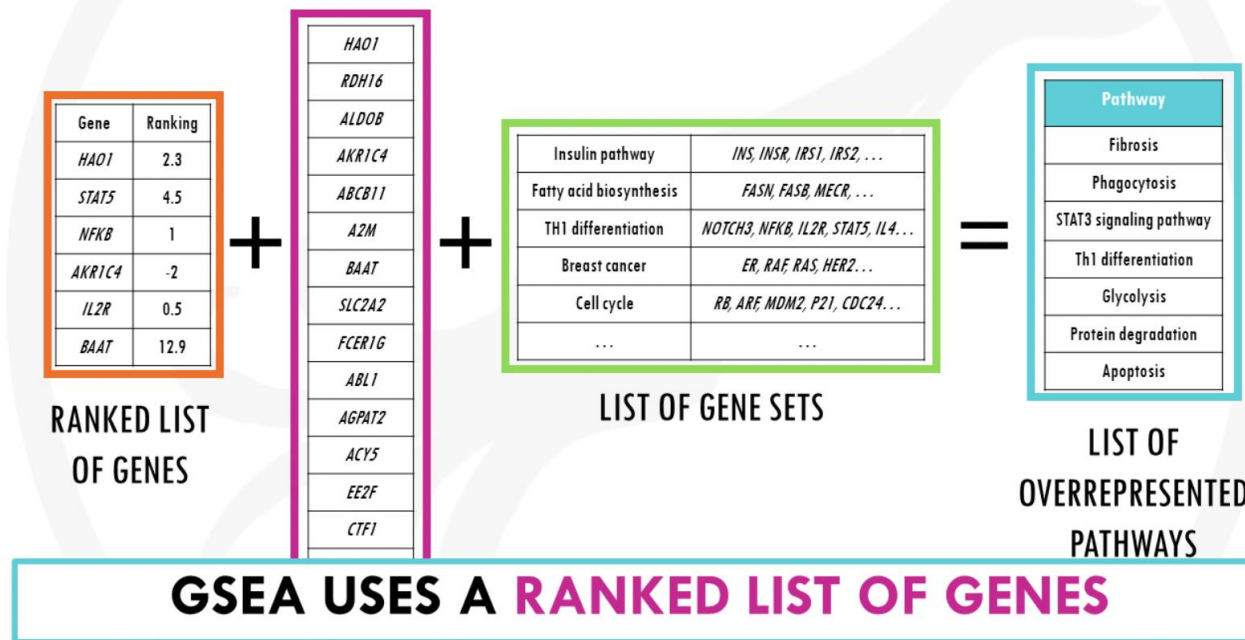
gene	sampleA	sampleB
EF2A	$1489 / 1.3 = \mathbf{1145.39}$	$906 / 0.77 = \mathbf{1176.62}$
ABCD1	$22 / 1.3 = \mathbf{16.92}$	$13 / 0.77 = \mathbf{16.88}$

Usually not performed “manually” as DESeq2 calculates it internally but we have to tell DESeq2 to do it

Functional analysis for biological interpretation

- **Gene-set enrichment analysis (GSEA)**

- **Purpose:** find biological terms (curated gene sets for a biological process) that are enriched in the data for biological interpretation
- **Input:** gene list ranked by fold-change (with sign) and p-value
- **Advantage over other overrepresentation methods:** considers direction of fold-change and statistical significance (no pre-filtering of genes necessary)



List of background genes
(for statistical tests)

Functional analysis for biological interpretation

- Gene-set enrichment analysis (GSEA) – ranked input gene list

List of all genes from differential gene expression analysis

	Gene symbol	p-val	Differentially expressed	FC
1	<i>HAO1</i>	0.0001	Yes	-5.6
2	<i>RDH16</i>	0.00045	Yes	-7.6
3	<i>ALDOB</i>	0.00009	Yes	-6.7
4	<i>AKR1C4</i>	0.0051	Yes	-10.0
5	<i>ABCB11</i>	0.067	Yes	10.0
6	<i>BAAT</i>	0.0034	Yes	23.5
7	<i>SLC2A2</i>	0.00990	Yes	2
8	<i>FCER1G</i>	0.00673	Yes	6.7
...				
6532	<i>DSM</i>	0.00009	Yes	9
6533	<i>VEGFA</i>	0.067	Yes	8.5
6534	<i>ZAP70</i>	0.0034	Yes	7.8

$$\text{RANKING} = \text{SIGN}(\text{FC}) * -\log_{10}(\text{p-value})$$

Ranked list of genes

	Gene symbol	Ranking
1	<i>JAK9</i>	12.9
2	<i>RDH16</i>	11.4
3	<i>ALDOB</i>	11.3
5	<i>AKR1C4</i>	10.7
6	<i>ABCB11</i>	8.6
7	<i>BAAT</i>	3.6
8	<i>SLC2A2</i>	0.0099
9	<i>FCER1G</i>	0.00673
	...	
6532	<i>MyD88</i>	-9
6533	<i>TRIF</i>	-9.37
6534	<i>IRAK1</i>	-15.6

SIGNIFICANT + UPREGULATED

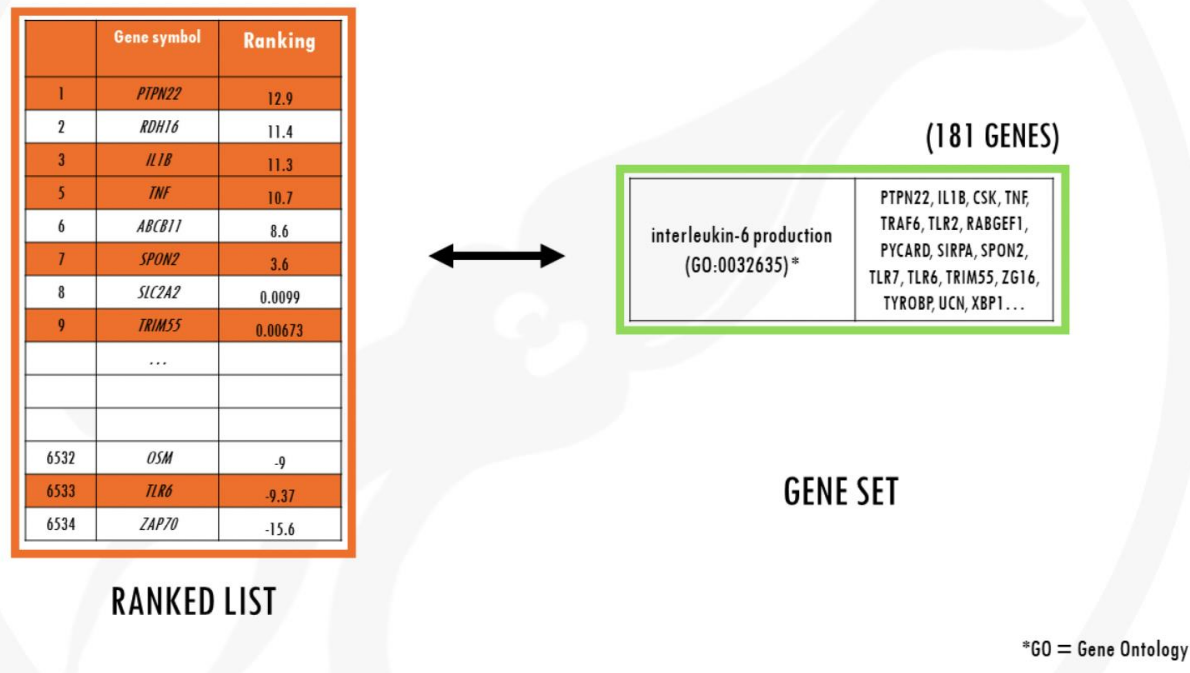
NON-SIGNIFICANT

SIGNIFICANT + DOWNREGULATED

Functional analysis for biological interpretation

- **Gene-set enrichment analysis (GSEA) – Which biological processes are enriched at the top and at the bottom of the ranked gene list?**

Result will show genes and biological terms that are



Up-regulated

[illegible]

Down-regulated

[illegible]

Not
enriched

[illegible]

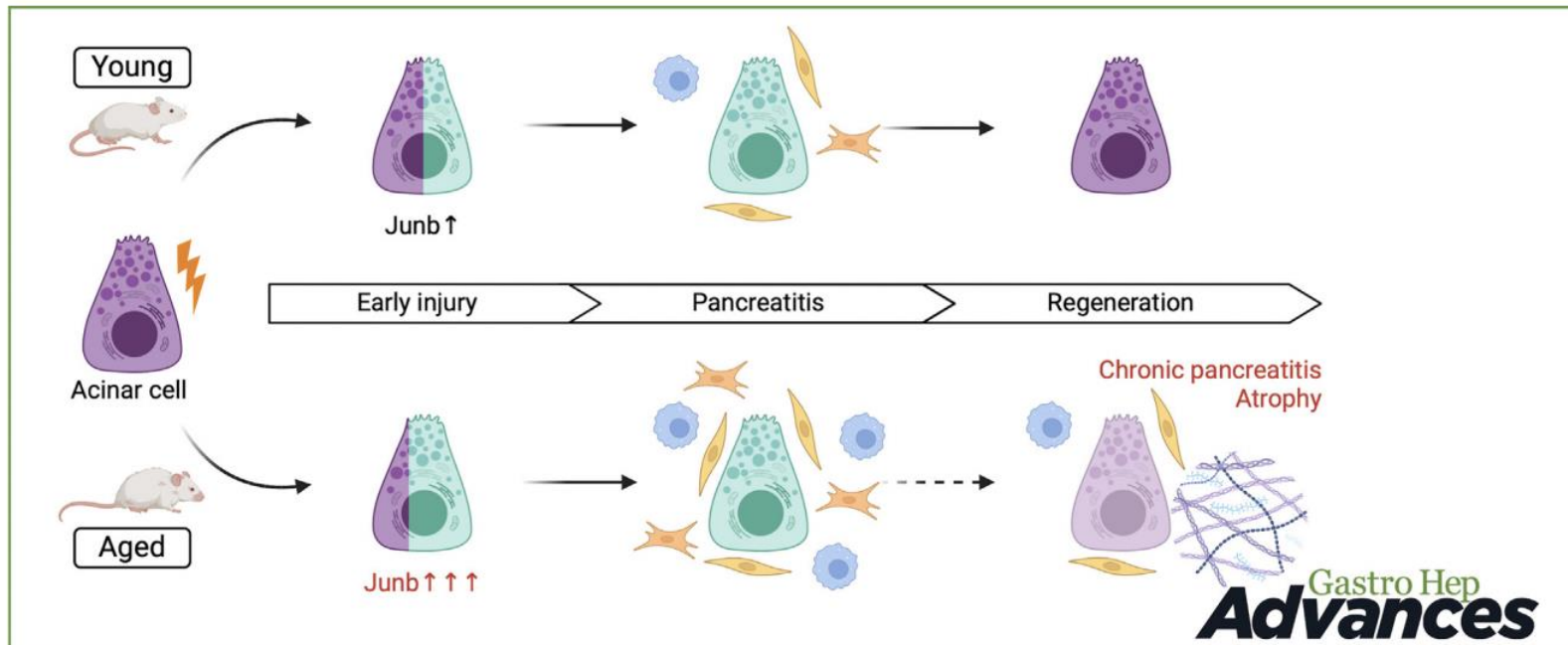
Hands-on data analysis: case study used in this course

Age-Related Decline in Pancreas Regeneration Is Associated With an Increased Proinflammatory Response to Injury



Kristina Høj,¹ Jonathan Baldan,^{1,†} Philip Allan Seymour,¹ Charlotte Vestrup Rift,² Jane Preuss Hasselby,² Albin Sandelin,^{1,3} and Luis Arnes¹

¹Biotech Research and Innovation Centre (BRIC), University of Copenhagen, Copenhagen, Denmark; ²Department of Pathology, Copenhagen University Hospital Rigshospitalet, Copenhagen, Denmark; and ³Department of Biology, University of Copenhagen Copenhagen, Denmark

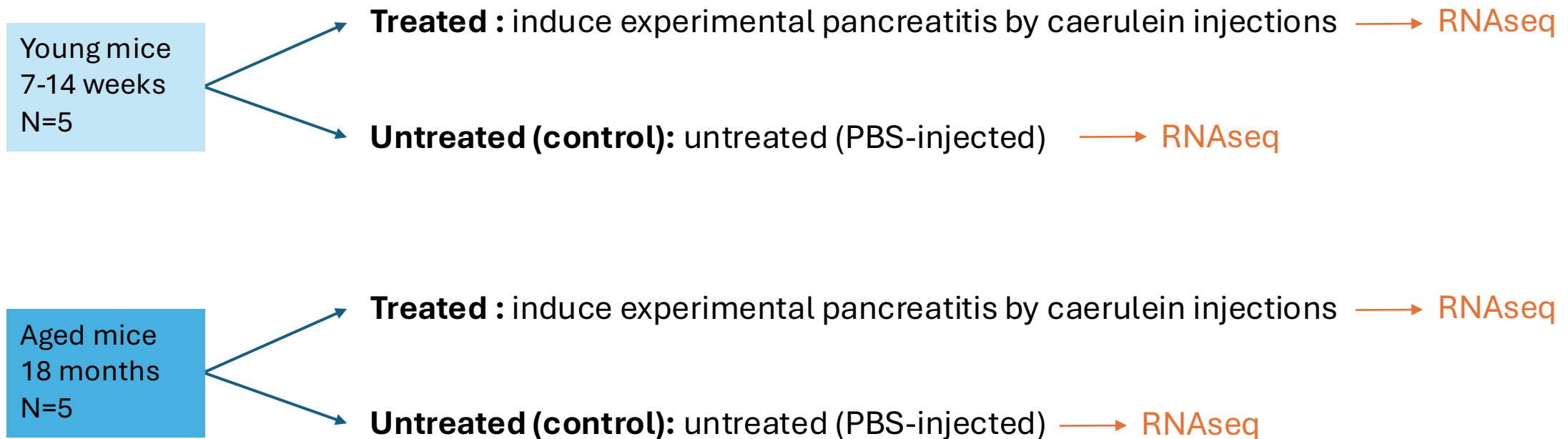


BACKGROUND AND AIMS

- The regenerative capacity of the pancreas diminishes with age.
- Understanding acinar cell responses to injury and the resolution of regenerative processes is crucial for tissue homeostasis.
- Knowledge about the impact of aging on these processes remains limited.

Case study used in this course – study design

To investigate the influence of aging on pancreas regeneration, the following study design was established.



- Question 1:** What is the **effect of treatment in young** mice? → young(treated vs control)
- Question 2:** What is the **effect of treatment in aged** mice? → aged(treated vs control)
- Question 3:** What is the difference between **aged vs young** under **control** conditions (no treatment)? → aged(control) vs young(control)
- Question 4:** What is the difference between **aged vs young** under **treatment**? → aged(treated) vs young(treated)
- Question 5:** Does **treatment affect aged and young differently** (interaction only)? → aged(treated vs control) vs young(treated vs control)

RNAseq data analysis hands-on

- R scripts in codspace (Juliana)

3. Results reporting using vuegen

- Hands-on demo (Albert)

Depositing RNAseq raw data and metadata

- For publications, RNAseq data have to be deposited in public archives
- Data will be assigned unique accession numbers for locating projects and data retrieval
- **Gene Expression Omnibus (GEO)** is the standard repository to deposit expression data: <https://www.ncbi.nlm.nih.gov/geo/>
- Data is also exchanged with the **European Nucleotide Archive (ENA)** from the European Bioinformatics Institute: <https://www.ebi.ac.uk/ena/browser/home>



Wrap up and final Q&A

Thanks for attending!

Feel free to contact us for help in RNAseq or other omics processing and analyses projects!

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